



Fpt Software

Nguyen Minh Chi

AI Engineer | NLP | AI Agents

SUMMARY

AI Engineer focused on AI agent engineering, with 3+ years of experience building agentic AI systems, retrieval-augmented generation (RAG), LLM-powered applications, and intelligent document processing. Strong foundation in machine learning theory and system design, with hands-on experience translating research concepts into FastAPI-based services and reliable production-oriented systems.

- **Agent Orchestration:** Design LangGraph-based and multi-agent workflows with routing, retrieval, validation, guardrails, and self-correction loops.
- **RAG & Vector Retrieval:** Build RAG pipelines with ChromaDB/FAISS-style vector retrieval, metadata enrichment, grounding accuracy, and low-latency responses.
- **API & LLM Integration:** Develop FastAPI services, integrate LLM APIs such as OpenAI, and build asynchronous LLM/OCR inference workflows.
- **Prompting & Validation:** Apply prompt engineering, task-specific evaluation workflows, and output validation patterns for reliable AI responses.

WORKING EXPERIENCE

1. Personal Project - VinBigData

Project: Life Assistant: Persistent Multi-Agent System

Jan 2026 – April 2026

Designed and implemented a persistent multi-agent life assistant with LangGraph-based orchestration, long-term memory, intent routing, RAG-based context retrieval, and real-time web research to deliver personalized, context-aware decisions.

- **Responsibilities:** Built a modular LangGraph architecture with specialized nodes for intent detection, memory retrieval, web research, response generation, Pydantic-based structured outputs, and critique-based self-correction.
- **Technologies:** Python, LangGraph, Mem0, OpenAI, Tavily Search, FastAPI, Pydantic

2. AI Engineer Training Program by VinGroup – VinBigData

Selected as one of the top 70 candidates from 900+ applicants for a competitive AI training program covering AI fundamentals, applied labs, internal project rotations, and agentic AI applications.

- **Responsibilities:** Implemented ML/DL models through structured labs covering supervised learning, neural networks, and deep learning optimization; contributed to conversational AI systems and multi-agent enterprise automation workflows.
- **Technologies:** Linear Algebra, Statistics and Probability, Machine Learning, Computer Vision, NLP, Generative AI, AI Ethics

3. Smart Green Transformation Center – Vin University

Project: EV SHIFT

Dec 2025 – Mar 2026

Built data collection and quality assurance pipelines for domain-specific datasets, including crawling, preprocessing, validation, and semantic consistency checks.

- **Responsibilities:** Built AI-focused crawling pipelines with Playwright and Selenium; integrated Crawl4AI with LLMs for document processing, content quality scoring, prompt engineering, task-specific generation, and evaluation workflows.
- **Technologies:** Playwright, Selenium, VN_nlp_core, Crawl4AI, Matplotlib

4. IMT Solution – Vietnam

Project: Intelligent Helpdesk Chatbot

Mar 2025 – Jul 2025

Architected and deployed a production-ready Retrieval-Augmented Generation (RAG) chatbot designed to automate internal technical support and knowledge retrieval.

- **Responsibilities:** Engineered a complex RAG pipeline using **LangChain (LCEL)** and **Qdrant** for high-precision semantic search over enterprise knowledge bases; implemented conversational retrieval by condensing multi-turn chat history into standalone context-aware queries; developed a multimodal retrieval system supporting both technical documentation and image references; built a scalable backend with **FastAPI (LangServe)** and a responsive **Streamlit** frontend with real-time streaming; integrated **Langfuse** for end-to-end observability, tracing, and prompt cost monitoring.
- **Technologies:** LangChain, Qdrant, OpenAI, FastAPI, Docker, Langfuse, Azure DevOps

5. IMT Solution – Vietnam

Project: Intelligent Document Processing

Sep 2024 – Jan 2025

Designed and owned an end-to-end Intelligent Document Processing (IDP) proof of concept, integrating OCR and OpenAI-powered LLM inference stages through an asynchronous FastAPI backend.

- **Responsibilities:** Built the FastAPI API layer to parallelize OCR and LLM inference stages, reduced multi-page document latency by approximately 70%, evaluated OCR/multimodal LLM options, and delivered a PoC processing 2,000+ invoices per day with 95%+ field-level accuracy.
- **Technologies:** Azure DevOps, OpenAI, FastAPI, LangChain

6. Ton Duc Thang University

Project: Agentic Legal QA System

March 2024 – June 2024

Designed and implemented a multi-agent legal QA system with routing, RAG retrieval, answer validation, and guardrails, serving as the primary technical owner of the system architecture and evaluation framework.

- **Responsibilities:** Designed dual-layer RAG with structure-aware legal chunking, enriched metadata for vector retrieval, built validation/guardrail agents, fine-tuned Llama 3 8B and Mistral 7B with QLoRA, and evaluated against a single-step RAG baseline.
- **Technologies:** LangChain, LangGraph, OpenAI, Unsloth, Transformers, ChromaDB, RAGAS, Docling

TECHNICAL SKILLS

- **Languages:** Python, SQL, JavaScript
- **AI Agent Engineering:** Multi-agent systems, agent orchestration, routing, memory, output validation, guardrails, self-correction loops
- **Frameworks & APIs:** FastAPI, LangGraph, LangChain, LlamaIndex, Mem0, PyTorch, TensorFlow, vLLM, OpenAI API
- **RAG & Data:** ChromaDB, FAISS, MySQL, retrieval pipelines, metadata enrichment, dataset engineering, semantic quality checks
- **Prompt Engineering:** System prompts, task-specific prompts, data generation prompts, evaluation workflows
- **Soft Skills:** Time management, problem solving, critical thinking, resilience, decision making, interpersonal communication

EDUCATION

- **B.Sc. in Computer Science** – Ton Duc Thang University

GPA: 8.31/10

CERTIFICATES

- **Deep Learning Specialization** – Coursera, 2023
- **Natural Language Processing Specialization** – Coursera, 2023
- **TensorFlow Developer Professional Certificate** – Coursera, 2023
- **Generative AI with Large Language Models** – DeepLearning.AI, 2024
- **IELTS 7.0** – English proficiency, 2026

LANGUAGES

- **English:** General professional proficiency

Vietnamese: Native